

# QFT Notes

Gytis Vėjelis

2026

## Contents

|          |                                                             |           |
|----------|-------------------------------------------------------------|-----------|
| <b>1</b> | <b>Preliminaries</b>                                        | <b>4</b>  |
| 1.1      | Conventions and Units . . . . .                             | 4         |
| 1.2      | Indices and Metric . . . . .                                | 4         |
| 1.3      | Lorentz Transformations . . . . .                           | 4         |
| 1.4      | Momentum, On-shell Condition . . . . .                      | 4         |
| 1.5      | Fourier Transform Conventions . . . . .                     | 5         |
| 1.6      | Delta Functions . . . . .                                   | 5         |
| 1.7      | Distributions and the $i\epsilon$ Prescription . . . . .    | 5         |
| 1.8      | Heaviside Step Function . . . . .                           | 5         |
| 1.9      | Useful Integrals and Identities . . . . .                   | 6         |
| 1.10     | Notation Summary . . . . .                                  | 6         |
| <b>2</b> | <b>Classical Field Theory</b>                               | <b>7</b>  |
| 2.1      | Fields and the Action Principle . . . . .                   | 7         |
| 2.2      | Euler–Lagrange Equations for Fields . . . . .               | 7         |
| 2.3      | Canonical Momentum and Hamiltonian Density . . . . .        | 7         |
| 2.4      | Lorentz Invariance . . . . .                                | 8         |
| 2.5      | Spacetime Translations and Energy–Momentum Tensor . . . . . | 8         |
| 2.6      | Noether’s Theorem (General Form) . . . . .                  | 9         |
| 2.7      | Internal Symmetry Example: Global $U(1)$ . . . . .          | 9         |
| 2.8      | Rotations and Angular Momentum (Useful Formulas) . . . . .  | 9         |
| 2.9      | Classical Poisson Brackets for Fields . . . . .             | 10        |
| 2.10     | Useful Lagrangians (Most Common) . . . . .                  | 10        |
| 2.11     | Dimensional Analysis (Quick Exam Tool) . . . . .            | 10        |
| 2.12     | Summary (What to Remember) . . . . .                        | 10        |
| <b>3</b> | <b>Canonical Quantization</b>                               | <b>11</b> |
| 3.1      | Klein–Gordon Equation (Classical) . . . . .                 | 11        |
| 3.2      | Warm-up: Simple Harmonic Oscillator (SHO) . . . . .         | 11        |
| 3.3      | Canonical Quantization of a Real Scalar Field . . . . .     | 11        |
| 3.4      | Mode Expansion and Free Quantum Field . . . . .             | 11        |
| 3.5      | Hamiltonian and Vacuum Energy . . . . .                     | 12        |
| 3.6      | Vacuum and Particles . . . . .                              | 12        |
| 3.7      | Relativistic Normalization . . . . .                        | 12        |
| 3.8      | Complex Scalar Fields . . . . .                             | 13        |
| 3.9      | The Heisenberg Picture . . . . .                            | 13        |
| 3.10     | Causality and Microcausality . . . . .                      | 13        |
| 3.11     | Propagators and Green’s Functions . . . . .                 | 14        |
| 3.12     | Applications (What Propagators Are For) . . . . .           | 14        |
| 3.13     | Non-Relativistic Field Theory (Basics) . . . . .            | 14        |
| 3.14     | Summary (Exam Checklist) . . . . .                          | 15        |

|          |                                                          |           |
|----------|----------------------------------------------------------|-----------|
| <b>4</b> | <b>Interacting Fields</b>                                | <b>16</b> |
| 4.1      | Types of Interaction (Common Examples)                   | 16        |
| 4.2      | The Interaction Picture                                  | 16        |
| 4.3      | Dyson's Formula (Dyson Series)                           | 16        |
| 4.4      | Scattering and the $S$ -Matrix                           | 17        |
| 4.5      | Wick's Theorem                                           | 17        |
| 4.6      | Feynman Diagrams (Meaning)                               | 17        |
| 4.7      | Feynman Rules (Scalar $\phi^4$ Theory)                   | 18        |
| 4.8      | Amplitudes and the Invariant Matrix Element              | 18        |
| 4.9      | Decay Rates and Cross Sections (Core Formulas)           | 18        |
| 4.10     | Green's Functions (Correlation Functions)                | 19        |
| 4.11     | Connected Diagrams and Vacuum Bubbles                    | 19        |
| 4.12     | Reduction Formula (LSZ)                                  | 19        |
| 4.13     | Summary (Exam Checklist)                                 | 20        |
| <b>5</b> | <b>The Dirac Equation</b>                                | <b>21</b> |
| 5.1      | The Lorentz Group (Quick Facts)                          | 21        |
| 5.2      | Clifford Algebra and Gamma Matrices                      | 21        |
| 5.3      | Spinor Representation of Lorentz Transformations         | 21        |
| 5.4      | The Dirac Equation and Dirac Lagrangian                  | 21        |
| 5.5      | Chiral Projectors and Chiral Spinors                     | 22        |
| 5.6      | The Weyl Equation (Massless Spinors)                     | 22        |
| 5.7      | Parity                                                   | 22        |
| 5.8      | Majorana Spinors (Idea)                                  | 22        |
| 5.9      | Symmetries and Conserved Currents                        | 23        |
| 5.10     | Plane Wave Solutions                                     | 23        |
| 5.11     | Summary (Exam Checklist)                                 | 24        |
| <b>6</b> | <b>Quantizing the Dirac Field</b>                        | <b>25</b> |
| 6.1      | A Glimpse at the Spin-Statistics Theorem (What You Need) | 25        |
| 6.2      | Canonical Fermionic Quantization                         | 25        |
| 6.3      | Mode Expansion of the Dirac Field                        | 25        |
| 6.4      | Fermi-Dirac Statistics and the Pauli Principle           | 26        |
| 6.5      | Hamiltonian and Energy Positivity                        | 26        |
| 6.6      | Particles and Antiparticles                              | 26        |
| 6.7      | Dirac's Hole Interpretation (Historical Motivation)      | 26        |
| 6.8      | Fermion Propagator                                       | 27        |
| 6.9      | Spin Sums (Used Constantly)                              | 27        |
| 6.10     | Wick's Theorem for Fermions (Sign Rules)                 | 27        |
| 6.11     | Feynman Rules (Dirac Fermions, Minimal Set)              | 27        |
| 6.12     | Summary (Exam Checklist)                                 | 28        |
| <b>7</b> | <b>Quantum Electrodynamics (QED)</b>                     | <b>29</b> |
| 7.1      | Gauge Invariance (Classical EM)                          | 29        |
| 7.2      | Inclusion of Matter: Minimal Coupling (QED Lagrangian)   | 29        |
| 7.3      | Conserved Current                                        | 29        |
| 7.4      | Quantization of the EM Field (Key Issue)                 | 30        |
| 7.5      | Lorentz-Invariant Propagators                            | 30        |

|       |                                                                       |    |
|-------|-----------------------------------------------------------------------|----|
| 7.6   | QED Feynman Rules (Momentum Space) . . . . .                          | 30 |
| 7.7   | QED Processes (Core Examples) . . . . .                               | 31 |
| 7.7.1 | Electron–Muon Scattering: $e^- \mu^- \rightarrow e^- \mu^-$ . . . . . | 31 |
| 7.7.2 | $e^+ e^- \rightarrow \mu^+ \mu^-$ (Annihilation) . . . . .            | 31 |
| 7.7.3 | Compton Scattering: $\gamma e^- \rightarrow \gamma e^-$ . . . . .     | 31 |
| 7.7.4 | Møller and Bhabha Scattering . . . . .                                | 31 |
| 7.8   | Gauge Invariance and Ward Identity (Essential Idea) . . . . .         | 32 |
| 7.9   | Summary (Exam Checklist) . . . . .                                    | 32 |

## 1 Preliminaries

### 1.1 Conventions and Units

We use natural units:

$$\hbar = c = 1. \quad (1)$$

Mass, energy, momentum have the same dimensions:

$$[E] = [p] = [m] = L^{-1}. \quad (2)$$

### 1.2 Indices and Metric

Spacetime coordinates:

$$x^\mu = (t, \mathbf{x}), \quad \mu = 0, 1, 2, 3. \quad (3)$$

Metric signature (choose one and be consistent). Common choice:

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1). \quad (4)$$

Scalar product:

$$x \cdot y := x^\mu y_\mu = x^0 y^0 - \mathbf{x} \cdot \mathbf{y}. \quad (5)$$

Raising/lowering indices:

$$x_\mu = \eta_{\mu\nu} x^\nu. \quad (6)$$

Derivatives:

$$\partial_\mu := \frac{\partial}{\partial x^\mu}, \quad \partial^\mu = \eta^{\mu\nu} \partial_\nu. \quad (7)$$

D'Alembertian:

$$\square := \partial_\mu \partial^\mu = \partial_t^2 - \nabla^2. \quad (8)$$

### 1.3 Lorentz Transformations

Lorentz transformation:

$$x'^\mu = \Lambda^\mu_\nu x^\nu, \quad (9)$$

with invariance condition:

$$\eta_{\rho\sigma} \Lambda^\rho_\mu \Lambda^\sigma_\nu = \eta_{\mu\nu}. \quad (10)$$

Invariant interval:

$$x^2 = x_\mu x^\mu = t^2 - |\mathbf{x}|^2. \quad (11)$$

### 1.4 Momentum, On-shell Condition

Four-momentum:

$$p^\mu = (E, \mathbf{p}), \quad p^2 = p_\mu p^\mu = E^2 - |\mathbf{p}|^2. \quad (12)$$

Mass shell condition:

$$p^2 = m^2 \quad \Rightarrow \quad E = \pm \sqrt{|\mathbf{p}|^2 + m^2}. \quad (13)$$

Usually take positive energy:

$$E_{\mathbf{p}} := \sqrt{|\mathbf{p}|^2 + m^2}. \quad (14)$$

## 1.5 Fourier Transform Conventions

A common convention in QFT:

$$f(x) = \int \frac{d^4 p}{(2\pi)^4} e^{-ip \cdot x} \tilde{f}(p), \quad (15)$$

$$\tilde{f}(p) = \int d^4 x e^{ip \cdot x} f(x). \quad (16)$$

Spatial Fourier transform:

$$f(\mathbf{x}) = \int \frac{d^3 p}{(2\pi)^3} e^{i\mathbf{p} \cdot \mathbf{x}} \tilde{f}(\mathbf{p}), \quad (17)$$

$$\tilde{f}(\mathbf{p}) = \int d^3 x e^{-i\mathbf{p} \cdot \mathbf{x}} f(\mathbf{x}). \quad (18)$$

Useful identity:

$$\int d^4 x e^{i(p-q) \cdot x} = (2\pi)^4 \delta^{(4)}(p - q). \quad (19)$$

## 1.6 Delta Functions

Dirac delta (1D):

$$\int_{-\infty}^{\infty} dx \delta(x - a) f(x) = f(a). \quad (20)$$

Scaling rule:

$$\delta(ax) = \frac{1}{|a|} \delta(x). \quad (21)$$

Multi-dimensional delta:

$$\delta^{(n)}(\mathbf{x}) = \prod_{i=1}^n \delta(x_i), \quad \int d^n x \delta^{(n)}(\mathbf{x} - \mathbf{a}) f(\mathbf{x}) = f(\mathbf{a}). \quad (22)$$

## 1.7 Distributions and the $i\epsilon$ Prescription

A key distribution identity:

$$\frac{1}{x \pm i\epsilon} = \mathcal{P} \left( \frac{1}{x} \right) \mp i\pi \delta(x), \quad \epsilon \rightarrow 0^+. \quad (23)$$

Here  $\mathcal{P}$  denotes the Cauchy principal value.

## 1.8 Heaviside Step Function

Definition:

$$\theta(t) = \begin{cases} 1, & t > 0, \\ 0, & t < 0. \end{cases} \quad (24)$$

Distribution derivative:

$$\frac{d}{dt} \theta(t) = \delta(t). \quad (25)$$

## 1.9 Useful Integrals and Identities

Gaussian integral (real,  $a > 0$ ):

$$\int_{-\infty}^{\infty} dx e^{-ax^2} = \sqrt{\frac{\pi}{a}}. \quad (26)$$

Integration by parts (fields vanish at infinity):

$$\int d^4x (\partial_\mu f) g = - \int d^4x f (\partial_\mu g). \quad (27)$$

## 1.10 Notation Summary

$$x^\mu = (t, \mathbf{x}), \quad p^\mu = (E, \mathbf{p}), \quad (28)$$

$$x \cdot p = x^\mu p_\mu = Et - \mathbf{p} \cdot \mathbf{x}, \quad \square = \partial_\mu \partial^\mu. \quad (29)$$

## 2 Classical Field Theory

### 2.1 Fields and the Action Principle

A (classical) field is a function on spacetime:

$$\phi_a(x), \quad x^\mu = (t, \mathbf{x}), \quad a = 1, \dots, N, \quad (30)$$

where  $a$  labels different field components (e.g. internal indices).

The dynamics is determined by an action functional

$$S[\phi] = \int d^4x \mathcal{L}(\phi_a, \partial_\mu \phi_a, x), \quad (31)$$

where  $\mathcal{L}$  is the *Lagrangian density*. For local relativistic theories,

$$\mathcal{L} = \mathcal{L}(\phi_a, \partial_\mu \phi_a), \quad (32)$$

i.e. no explicit  $x$  dependence.

### 2.2 Euler–Lagrange Equations for Fields

Vary the action under  $\phi_a \rightarrow \phi_a + \delta\phi_a$  with  $\delta\phi_a$  vanishing on the boundary:

$$\delta S = 0 \quad \Rightarrow \quad \frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left( \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) = 0. \quad (33)$$

**Example: Klein–Gordon field.**

$$\mathcal{L}_{\text{KG}} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2. \quad (34)$$

Then

$$(\square + m^2)\phi = 0. \quad (35)$$

**Example: complex scalar.**

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi. \quad (36)$$

Treat  $\phi$  and  $\phi^*$  as independent in variation:

$$(\square + m^2)\phi = 0, \quad (\square + m^2)\phi^* = 0. \quad (37)$$

### 2.3 Canonical Momentum and Hamiltonian Density

Define canonical conjugate momentum fields:

$$\pi_a(x) := \frac{\partial \mathcal{L}}{\partial (\partial_0 \phi_a)}. \quad (38)$$

Hamiltonian density:

$$\mathcal{H} := \pi_a \dot{\phi}_a - \mathcal{L}, \quad \dot{\phi}_a := \partial_0 \phi_a. \quad (39)$$

Total Hamiltonian:

$$H = \int d^3x \mathcal{H}. \quad (40)$$

Hamilton's equations (field form):

$$\dot{\phi}_a(\mathbf{x}, t) = \frac{\delta H}{\delta \pi_a(\mathbf{x}, t)}, \quad \dot{\pi}_a(\mathbf{x}, t) = -\frac{\delta H}{\delta \phi_a(\mathbf{x}, t)}. \quad (41)$$

**Example: Klein–Gordon Hamiltonian.** For  $\mathcal{L}_{\text{KG}}$ :

$$\pi = \dot{\phi}, \quad (42)$$

and

$$\mathcal{H} = \frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2, \quad H = \int d^3x \left[ \frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2 \right]. \quad (43)$$

## 2.4 Lorentz Invariance

A theory is Lorentz invariant if the action is invariant under

$$x^\mu \rightarrow x'^\mu = \Lambda^\mu_\nu x^\nu, \quad (44)$$

with fields transforming in some representation:

$$\phi_a(x) \rightarrow \phi'_a(x') = D(\Lambda)_{ab} \phi_b(x). \quad (45)$$

The action is invariant if

$$S[\phi] = S[\phi']. \quad (46)$$

For a scalar field ( $D(\Lambda) = 1$ ):

$$\phi'(x') = \phi(x). \quad (47)$$

**Lorentz-covariant building blocks.** Common invariant terms in  $\mathcal{L}$ :

$$\phi^2, \quad \partial_\mu \phi \partial^\mu \phi, \quad \phi^* \phi, \quad \partial_\mu \phi^* \partial^\mu \phi. \quad (48)$$

## 2.5 Spacetime Translations and Energy–Momentum Tensor

For an infinitesimal translation

$$x^\mu \rightarrow x'^\mu = x^\mu + \epsilon^\mu, \quad (49)$$

the action is invariant if  $\mathcal{L}$  has no explicit  $x$  dependence.

The *canonical energy–momentum tensor* is

$$T^\mu_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \partial_\nu \phi_a - \delta^\mu_\nu \mathcal{L}. \quad (50)$$

Conservation law:

$$\partial_\mu T^\mu_\nu = 0. \quad (51)$$

Energy and momentum:

$$P^\nu = \int d^3x T^{0\nu}, \quad H = P^0 = \int d^3x T^{00}. \quad (52)$$

**Remark (symmetrization).** The canonical  $T^{\mu\nu}$  is not always symmetric. One can construct an improved symmetric tensor (Belinfante):

$$T_{\text{sym}}^{\mu\nu} = T^{\mu\nu} + \partial_\lambda B^{\lambda\mu\nu}, \quad (53)$$

where  $B^{\lambda\mu\nu}$  is antisymmetric in  $\lambda \leftrightarrow \mu$ , so  $\partial_\mu T_{\text{sym}}^{\mu\nu} = 0$  still holds.

## 2.6 Noether's Theorem (General Form)

Suppose the action is invariant under a continuous transformation with

$$\phi_a(x) \rightarrow \phi_a(x) + \delta\phi_a(x), \quad x^\mu \rightarrow x^\mu + \delta x^\mu, \quad (54)$$

such that the Lagrangian changes by a total derivative:

$$\delta\mathcal{L} = \partial_\mu K^\mu. \quad (55)$$

Then there exists a conserved current  $j^\mu$ :

$$\partial_\mu j^\mu = 0, \quad (56)$$

with conserved charge

$$Q = \int d^3x j^0, \quad \frac{dQ}{dt} = 0. \quad (57)$$

A useful expression for the Noether current is:

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)} \delta\phi_a + \mathcal{L} \delta x^\mu - K^\mu. \quad (58)$$

## 2.7 Internal Symmetry Example: Global $U(1)$

Complex scalar field with

$$\mathcal{L} = \partial_\mu\phi^* \partial^\mu\phi - m^2\phi^*\phi. \quad (59)$$

Global phase symmetry:

$$\phi \rightarrow e^{i\alpha}\phi, \quad \phi^* \rightarrow e^{-i\alpha}\phi^*, \quad \alpha = \text{const.} \quad (60)$$

Infinitesimal:

$$\delta\phi = i\alpha\phi, \quad \delta\phi^* = -i\alpha\phi^*. \quad (61)$$

Noether current:

$$j^\mu = i(\phi^* \partial^\mu\phi - (\partial^\mu\phi^*)\phi). \quad (62)$$

Conservation:

$$\partial_\mu j^\mu = 0. \quad (63)$$

Conserved charge:

$$Q = \int d^3x j^0. \quad (64)$$

## 2.8 Rotations and Angular Momentum (Useful Formulas)

Lorentz invariance implies conservation of angular momentum (including boosts). Define the Lorentz current:

$$M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu} + (\text{spin part, if field has spin}). \quad (65)$$

Conservation:

$$\partial_\lambda M^{\lambda\mu\nu} = 0. \quad (66)$$

Angular momentum generators:

$$J^{\mu\nu} = \int d^3x M^{0\mu\nu}. \quad (67)$$

## 2.9 Classical Poisson Brackets for Fields

At equal time  $t$ :

$$\{\phi_a(\mathbf{x}, t), \pi_b(\mathbf{y}, t)\} = \delta_{ab} \delta^{(3)}(\mathbf{x} - \mathbf{y}), \quad (68)$$

$$\{\phi_a(\mathbf{x}, t), \phi_b(\mathbf{y}, t)\} = 0, \quad \{\pi_a(\mathbf{x}, t), \pi_b(\mathbf{y}, t)\} = 0. \quad (69)$$

Time evolution of any functional  $F[\phi, \pi]$ :

$$\dot{F} = \{F, H\} + \frac{\partial F}{\partial t}. \quad (70)$$

## 2.10 Useful Lagrangians (Most Common)

Real scalar ( $\phi^4$  theory):

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{4!} \phi^4. \quad (71)$$

Complex scalar with interaction:

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi - \frac{\lambda}{4} (\phi^* \phi)^2. \quad (72)$$

## 2.11 Dimensional Analysis (Quick Exam Tool)

In  $d = 4$  spacetime dimensions:

$$[\mathcal{L}] = 4. \quad (73)$$

For a real scalar kinetic term:

$$\mathcal{L} \sim (\partial \phi)^2 \quad \Rightarrow \quad [\phi] = 1. \quad (74)$$

Then:

$$[m] = 1, \quad [\lambda_{\phi^4}] = 0. \quad (75)$$

So  $\phi^4$  coupling is dimensionless (renormalizable in 4d).

## 2.12 Summary (What to Remember)

- Field equations follow from  $\delta S = 0$ :

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left( \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) = 0.$$

- Canonical momentum:  $\pi_a = \partial \mathcal{L} / \partial \dot{\phi}_a$ .
- Hamiltonian density:  $\mathcal{H} = \pi_a \dot{\phi}_a - \mathcal{L}$ .
- Noether theorem: continuous symmetry  $\Rightarrow$  conserved current  $\partial_\mu j^\mu = 0$ .
- Translation invariance  $\Rightarrow$  conserved  $T^{\mu\nu}$  and  $P^\mu$ .

### 3 Canonical Quantization

#### 3.1 Klein–Gordon Equation (Classical)

For a real scalar field:

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \quad \Rightarrow \quad (\square + m^2) \phi = 0. \quad (76)$$

Plane-wave solutions:

$$\phi(x) \sim e^{-ip \cdot x}, \quad p^2 = m^2, \quad p^0 = \pm E_{\mathbf{p}}, \quad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}. \quad (77)$$

#### 3.2 Warm-up: Simple Harmonic Oscillator (SHO)

Hamiltonian:

$$H = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2. \quad (78)$$

Define ladder operators:

$$a = \sqrt{\frac{m\omega}{2}} \left( x + \frac{i}{m\omega} p \right), \quad a^\dagger = \sqrt{\frac{m\omega}{2}} \left( x - \frac{i}{m\omega} p \right), \quad (79)$$

with

$$[a, a^\dagger] = 1. \quad (80)$$

Then

$$H = \omega \left( a^\dagger a + \frac{1}{2} \right), \quad E_n = \omega \left( n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots \quad (81)$$

#### 3.3 Canonical Quantization of a Real Scalar Field

Promote  $\phi, \pi$  to operators and impose equal-time commutators:

$$\pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}, \quad (82)$$

$$[\phi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = i \delta^{(3)}(\mathbf{x} - \mathbf{y}), \quad [\phi(\mathbf{x}, t), \phi(\mathbf{y}, t)] = 0, \quad [\pi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = 0. \quad (83)$$

#### 3.4 Mode Expansion and Free Quantum Field

The general operator solution of the KG equation is expanded as:

$$\phi(x) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} (a_{\mathbf{p}} e^{-ip \cdot x} + a_{\mathbf{p}}^\dagger e^{ip \cdot x}), \quad p^0 = E_{\mathbf{p}}. \quad (84)$$

Imposing canonical commutators implies:

$$[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}), \quad [a_{\mathbf{p}}, a_{\mathbf{q}}] = 0, \quad [a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}^\dagger] = 0. \quad (85)$$

### 3.5 Hamiltonian and Vacuum Energy

For the free KG field,

$$\mathcal{H} = \frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2, \quad H = \int d^3x \mathcal{H}. \quad (86)$$

In terms of creation/annihilation operators:

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left( a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + \frac{1}{2}(2\pi)^3 \delta^{(3)}(0) \right). \quad (87)$$

The  $\delta^{(3)}(0)$  term corresponds to the (divergent) **vacuum energy**:

$$E_0 = \frac{1}{2} \sum_{\mathbf{p}} E_{\mathbf{p}} \rightarrow \infty. \quad (88)$$

**Normal ordering.** Define normal ordering by moving all creation operators to the left:

$$: H : = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} a_{\mathbf{p}}^\dagger a_{\mathbf{p}}, \quad (89)$$

so that

$$\langle 0 | : H : | 0 \rangle = 0. \quad (90)$$

### 3.6 Vacuum and Particles

Define the vacuum by

$$a_{\mathbf{p}} |0\rangle = 0 \quad \forall \mathbf{p}. \quad (91)$$

One-particle state:

$$|\mathbf{p}\rangle = a_{\mathbf{p}}^\dagger |0\rangle. \quad (92)$$

Multi-particle states:

$$|\mathbf{p}_1, \dots, \mathbf{p}_n\rangle = a_{\mathbf{p}_1}^\dagger \cdots a_{\mathbf{p}_n}^\dagger |0\rangle. \quad (93)$$

Number operator:

$$N = \int \frac{d^3p}{(2\pi)^3} a_{\mathbf{p}}^\dagger a_{\mathbf{p}}. \quad (94)$$

### 3.7 Relativistic Normalization

A common convention is

$$\langle \mathbf{p} | \mathbf{q} \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}). \quad (95)$$

A Lorentz-invariant measure is

$$\int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}}. \quad (96)$$

Covariant normalization often used in scattering:

$$|p\rangle_{\text{cov}} = \sqrt{2E_{\mathbf{p}}} |\mathbf{p}\rangle, \quad \langle p | q \rangle_{\text{cov}} = (2\pi)^3 2E_{\mathbf{p}} \delta^{(3)}(\mathbf{p} - \mathbf{q}). \quad (97)$$

### 3.8 Complex Scalar Fields

Lagrangian:

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi. \quad (98)$$

Canonical momenta:

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}^*, \quad \pi^* = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^*} = \dot{\phi}. \quad (99)$$

Mode expansions introduce **particles and antiparticles**:

$$\phi(x) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} (a_{\mathbf{p}} e^{-ip \cdot x} + b_{\mathbf{p}}^\dagger e^{ip \cdot x}), \quad (100)$$

$$\phi^*(x) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} (a_{\mathbf{p}}^\dagger e^{ip \cdot x} + b_{\mathbf{p}} e^{-ip \cdot x}). \quad (101)$$

Nonzero commutators:

$$[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}), \quad [b_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}). \quad (102)$$

**Conserved charge ( $U(1)$  symmetry).**

$$j^\mu = i(\phi^* \partial^\mu \phi - (\partial^\mu \phi^*) \phi), \quad Q = \int d^3 x j^0. \quad (103)$$

In terms of operators (schematically):

$$Q = \int \frac{d^3 p}{(2\pi)^3} (a_{\mathbf{p}}^\dagger a_{\mathbf{p}} - b_{\mathbf{p}}^\dagger b_{\mathbf{p}}). \quad (104)$$

### 3.9 The Heisenberg Picture

Operators evolve in time, states are time-independent:

$$\mathcal{O}_H(t) = e^{iHt} \mathcal{O}_S e^{-iHt}. \quad (105)$$

Equation of motion:

$$\frac{d}{dt} \mathcal{O}_H(t) = i[H, \mathcal{O}_H(t)] + \left( \frac{\partial \mathcal{O}}{\partial t} \right)_H. \quad (106)$$

For fields:

$$\dot{\phi}(\mathbf{x}, t) = i[H, \phi(\mathbf{x}, t)]. \quad (107)$$

### 3.10 Causality and Microcausality

Relativistic causality requires that measurements at spacelike separation commute:

$$(x - y)^2 < 0 \quad \Rightarrow \quad [\phi(x), \phi(y)] = 0. \quad (108)$$

Define the **Pauli–Jordan commutator function**:

$$[\phi(x), \phi(y)] = i\Delta(x - y), \quad (109)$$

where  $\Delta(x)$  vanishes for spacelike  $x$ .

### 3.11 Propagators and Green's Functions

The **Feynman propagator** is the time-ordered vacuum correlator:

$$\Delta_F(x-y) := \langle 0 | T \{ \phi(x) \phi(y) \} | 0 \rangle. \quad (110)$$

Time ordering:

$$T \{ \phi(x) \phi(y) \} = \theta(x^0 - y^0) \phi(x) \phi(y) + \theta(y^0 - x^0) \phi(y) \phi(x). \quad (111)$$

Momentum-space representation:

$$\Delta_F(x-y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{-ip \cdot (x-y)}. \quad (112)$$

It satisfies the Green's function equation:

$$(\square + m^2) \Delta_F(x-y) = -i \delta^{(4)}(x-y). \quad (113)$$

**Retarded/advanced propagators.** Retarded Green function:

$$(\square + m^2) \Delta_R(x) = -\delta^{(4)}(x), \quad \Delta_R(x) = 0 \text{ for } x^0 < 0. \quad (114)$$

Advanced Green function:

$$\Delta_A(x) = 0 \text{ for } x^0 > 0. \quad (115)$$

### 3.12 Applications (What Propagators Are For)

- They encode **free field correlations**:

$$\langle 0 | \phi(x) \phi(y) | 0 \rangle.$$

- They are Green functions used to solve sourced equations:

$$(\square + m^2) \phi(x) = J(x) \quad \Rightarrow \quad \phi(x) = \phi_{\text{hom}}(x) + \int d^4 y \Delta_R(x-y) J(y).$$

- In perturbation theory,  $\Delta_F$  becomes the internal line of Feynman diagrams.

### 3.13 Non-Relativistic Field Theory (Basics)

Non-relativistic quantum mechanics can be written in second-quantized form using a field  $\psi(\mathbf{x}, t)$  with Lagrangian

$$\mathcal{L} = i\psi^\dagger \partial_t \psi - \frac{1}{2m} \nabla \psi^\dagger \cdot \nabla \psi - V(\mathbf{x}) \psi^\dagger \psi. \quad (116)$$

Canonical momentum:

$$\pi_\psi = \frac{\partial \mathcal{L}}{\partial(\partial_t \psi)} = i\psi^\dagger. \quad (117)$$

Quantization (bosons):

$$[\psi(\mathbf{x}, t), \psi^\dagger(\mathbf{y}, t)] = \delta^{(3)}(\mathbf{x} - \mathbf{y}). \quad (118)$$

Hamiltonian:

$$H = \int d^3 x \left[ \frac{1}{2m} \nabla \psi^\dagger \cdot \nabla \psi + V(\mathbf{x}) \psi^\dagger \psi \right]. \quad (119)$$

Equation of motion (Heisenberg):

$$i\partial_t\psi(\mathbf{x}, t) = \left[ -\frac{\nabla^2}{2m} + V(\mathbf{x}) \right] \psi(\mathbf{x}, t), \quad (120)$$

which is the Schrödinger equation.

**Interactions (typical).** A common local interaction is

$$\mathcal{L}_{\text{int}} = -\frac{g}{2}(\psi^\dagger\psi)^2. \quad (121)$$

### 3.14 Summary (Exam Checklist)

- Canonical quantization:  $[\phi, \pi] = i\delta^{(3)}$  at equal time.
- Mode expansion introduces  $a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger$  and particles.
- Free Hamiltonian is a continuum of oscillators:

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left( a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + \frac{1}{2} \delta^{(3)}(0) \right).$$

- Vacuum energy diverges; normal ordering removes the constant shift.
- Complex scalar  $\Rightarrow$  particle/antiparticle operators  $a, b$  and conserved  $U(1)$  charge.
- Causality:  $[\phi(x), \phi(y)] = 0$  for spacelike separation.
- Feynman propagator:

$$\Delta_F(x) = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{-ip \cdot x}.$$

- Non-relativistic fields reproduce Schrödinger equation via second quantization.

## 4 Interacting Fields

### 4.1 Types of Interaction (Common Examples)

Interactions are added to the free Lagrangian:

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\text{int}}. \quad (122)$$

**Scalar interactions.**

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4, \quad \mathcal{L}_{\text{int}} = -\frac{g}{3!}\phi^3. \quad (123)$$

**Yukawa interaction (scalar + fermion).**

$$\mathcal{L}_{\text{int}} = -g \bar{\psi} \psi \phi. \quad (124)$$

**QED interaction.**

$$\mathcal{L}_{\text{int}} = -e \bar{\psi} \gamma^\mu \psi A_\mu. \quad (125)$$

**Key point.** The interaction is treated perturbatively when the coupling is small.

### 4.2 The Interaction Picture

Split the Hamiltonian:

$$H = H_0 + H_I, \quad (126)$$

where  $H_0$  is exactly solvable (free theory),  $H_I$  is the interaction.

Interaction-picture operators evolve with  $H_0$ :

$$\mathcal{O}_I(t) = e^{iH_0 t} \mathcal{O}_S e^{-iH_0 t}. \quad (127)$$

States evolve with  $H_I$ :

$$i \frac{d}{dt} |\psi(t)\rangle_I = H_I(t) |\psi(t)\rangle_I, \quad H_I(t) = e^{iH_0 t} H_I e^{-iH_0 t}. \quad (128)$$

Define the time-evolution operator in the interaction picture:

$$|\psi(t)\rangle_I = U_I(t, t_0) |\psi(t_0)\rangle_I, \quad i \frac{d}{dt} U_I(t, t_0) = H_I(t) U_I(t, t_0), \quad U_I(t_0, t_0) = \mathbf{1}. \quad (129)$$

### 4.3 Dyson's Formula (Dyson Series)

Formal solution:

$$U_I(t, t_0) = T \exp \left( -i \int_{t_0}^t dt' H_I(t') \right). \quad (130)$$

Expanding:

$$U_I(t, t_0) = \mathbf{1} - i \int_{t_0}^t dt_1 H_I(t_1) + \frac{(-i)^2}{2!} \int_{t_0}^t dt_1 dt_2 T \{ H_I(t_1) H_I(t_2) \} + \dots \quad (131)$$

In covariant form, for local interactions:

$$S \equiv U_I(\infty, -\infty) = T \exp \left( i \int d^4x \mathcal{L}_{\text{int}}(x) \right). \quad (132)$$

#### 4.4 Scattering and the $S$ -Matrix

Define asymptotic in/out states (free particle states):

$$|i\rangle = |\text{in}\rangle, \quad |f\rangle = |\text{out}\rangle. \quad (133)$$

The  $S$ -matrix maps in-states to out-states:

$$|f\rangle = S|i\rangle. \quad (134)$$

Matrix element:

$$S_{fi} = \langle f|S|i\rangle. \quad (135)$$

It is conventional to write

$$S = \mathbf{1} + iT, \quad (136)$$

so that the scattering amplitude is encoded in  $T$ .

**Momentum conservation.** For translational invariance,

$$\langle f|iT|i\rangle = i(2\pi)^4 \delta^{(4)}(p_f - p_i) \mathcal{M}_{fi}, \quad (137)$$

where  $\mathcal{M}$  is the invariant amplitude.

#### 4.5 Wick's Theorem

For free fields, time-ordered products can be expanded into normal-ordered products plus contractions.

For scalar fields:

$$T\{\phi(x)\phi(y)\} = :\phi(x)\phi(y): + \overline{\phi(x)\phi(y)}, \quad (138)$$

where the contraction is defined as

$$\overline{\phi(x)\phi(y)} \equiv \langle 0|T\{\phi(x)\phi(y)\}|0\rangle = \Delta_F(x - y). \quad (139)$$

General statement:

$$T\{\phi_1\phi_2\cdots\phi_n\} = :\phi_1\phi_2\cdots\phi_n: + (\text{all possible contractions}). \quad (140)$$

#### 4.6 Feynman Diagrams (Meaning)

In perturbation theory:

- Expand  $S$  in powers of the coupling.
- Use Wick's theorem to evaluate vacuum expectation values.
- Each term corresponds to a sum of diagrams.

Diagrams provide a bookkeeping method for contractions.

#### 4.7 Feynman Rules (Scalar $\phi^4$ Theory)

Take

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4. \quad (141)$$

Momentum-space Feynman rules:

- External line: attach momentum  $p$  (on-shell  $p^2 = m^2$ ).
- Propagator (internal line with momentum  $k$ ):

$$\frac{i}{k^2 - m^2 + i\epsilon}. \quad (142)$$

- Vertex ( $\phi^4$ ):

$$-i\lambda. \quad (143)$$

- Integrate over undetermined loop momenta:

$$\int \frac{d^4k}{(2\pi)^4}. \quad (144)$$

- Impose momentum conservation at each vertex:

$$(2\pi)^4 \delta^{(4)}\left(\sum p_{\text{in}} - \sum p_{\text{out}}\right). \quad (145)$$

- Divide by symmetry factors when needed.

#### 4.8 Amplitudes and the Invariant Matrix Element

For  $2 \rightarrow 2$  scattering:

$$\langle p_3, p_4 | iT | p_1, p_2 \rangle = i(2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4) \mathcal{M}. \quad (146)$$

**Tree-level example in  $\phi^4$ .** At lowest order,

$$\mathcal{M}_{\text{tree}} = -\lambda. \quad (147)$$

#### 4.9 Decay Rates and Cross Sections (Core Formulas)

**Lorentz-invariant phase space.** For a final state of  $n$  particles:

$$d\Pi_n = (2\pi)^4 \delta^{(4)}\left(p_i - \sum_{j=1}^n p_j\right) \prod_{j=1}^n \frac{d^3p_j}{(2\pi)^3 2E_{p_j}}. \quad (148)$$

**Decay rate.** For  $A \rightarrow 1 + \dots + n$ :

$$d\Gamma = \frac{1}{2m_A} |\mathcal{M}|^2 d\Pi_n. \quad (149)$$

**Cross section.** For  $2 \rightarrow n$ :

$$d\sigma = \frac{1}{4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}} |\mathcal{M}|^2 d\Pi_n. \quad (150)$$

In the center-of-mass (COM) frame, the flux factor simplifies to

$$4E_1 E_2 |\mathbf{v}_1 - \mathbf{v}_2|. \quad (151)$$

#### 4.10 Green's Functions (Correlation Functions)

Define the  $n$ -point time-ordered Green function:

$$G^{(n)}(x_1, \dots, x_n) = \langle 0 | T \{ \phi(x_1) \cdots \phi(x_n) \} | 0 \rangle. \quad (152)$$

These are the central objects in perturbation theory.

#### 4.11 Connected Diagrams and Vacuum Bubbles

The vacuum-to-vacuum amplitude:

$$\langle 0 | S | 0 \rangle = \langle 0 | T \exp \left( i \int d^4 x \mathcal{L}_{\text{int}}(x) \right) | 0 \rangle. \quad (153)$$

Diagrams with no external legs are **vacuum bubbles**. They factorize and cancel in properly normalized correlation functions:

$$\langle \Omega | T \{ \phi(x_1) \cdots \phi(x_n) \} | \Omega \rangle = \frac{\langle 0 | T \{ \phi(x_1) \cdots \phi(x_n) S \} | 0 \rangle}{\langle 0 | S | 0 \rangle}. \quad (154)$$

**Connected vs disconnected.** The generating functional (in path integral language) satisfies:

$$Z[J] = e^{iW[J]}, \quad (155)$$

where  $W[J]$  generates **connected** diagrams. (You may be expected to know this identity even in a canonical approach.)

#### 4.12 Reduction Formula (LSZ)

The LSZ reduction formula relates scattering amplitudes to time-ordered correlators.

For a real scalar field, schematically:

$$\langle p'_1, \dots, p'_m | S | p_1, \dots, p_n \rangle \propto \prod_{i=1}^m \left[ \int d^4 x_i e^{ip'_i \cdot x_i} (\square_{x_i} + m^2) \right] \prod_{j=1}^n \left[ \int d^4 y_j e^{-ip_j \cdot y_j} (\square_{y_j} + m^2) \right] \langle 0 | T \{ \phi(x_1) \cdots \phi(y_n) \} | 0 \rangle. \quad (156)$$

Interpretation:

- Apply  $(\square + m^2)$  to *amputate* external propagators.
- Fourier transform to put external legs on-shell.
- The result gives the  $S$ -matrix element (up to normalization factors).

### 4.13 Summary (Exam Checklist)

- Interaction picture: operators evolve with  $H_0$ , states with  $H_I$ .
- Dyson formula:

$$S = T \exp \left( i \int d^4x \mathcal{L}_{\text{int}}(x) \right).$$

- Wick theorem reduces time-ordered products to contractions  $\Delta_F$ .
- Feynman rules: propagator  $i/(p^2 - m^2 + i\epsilon)$ , vertices from  $\mathcal{L}_{\text{int}}$ .
- Scattering:

$$\langle f | iT | i \rangle = i(2\pi)^4 \delta^{(4)}(p_f - p_i) \mathcal{M}.$$

- Decays and cross sections computed from  $|\mathcal{M}|^2$  and phase space.
- Vacuum bubbles cancel in normalized correlators.
- LSZ relates  $S$ -matrix elements to amputated time-ordered Green functions.

## 5 The Dirac Equation

### 5.1 The Lorentz Group (Quick Facts)

Lorentz transformations preserve the Minkowski metric:

$$\eta_{\rho\sigma}\Lambda^\rho_\mu\Lambda^\sigma_\nu = \eta_{\mu\nu}. \quad (157)$$

Infinitesimally,

$$\Lambda^\mu_\nu = \delta^\mu_\nu + \omega^\mu_\nu, \quad \omega_{\mu\nu} = -\omega_{\nu\mu}. \quad (158)$$

Generators in the vector representation satisfy the Lorentz algebra:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(\eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - \eta^{\nu\sigma}J^{\mu\rho} + \eta^{\mu\sigma}J^{\nu\rho}). \quad (159)$$

### 5.2 Clifford Algebra and Gamma Matrices

The Dirac gamma matrices  $\gamma^\mu$  satisfy the Clifford algebra:

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}. \quad (160)$$

Define

$$\gamma^5 := i\gamma^0\gamma^1\gamma^2\gamma^3, \quad (161)$$

which obeys

$$(\gamma^5)^2 = \mathbf{1}, \quad \{\gamma^5, \gamma^\mu\} = 0. \quad (162)$$

Useful matrices:

$$\sigma^{\mu\nu} := \frac{i}{2}[\gamma^\mu, \gamma^\nu]. \quad (163)$$

### 5.3 Spinor Representation of Lorentz Transformations

Dirac spinors transform as

$$\psi(x) \rightarrow \psi'(x') = S(\Lambda)\psi(x), \quad x' = \Lambda x, \quad (164)$$

with

$$S(\Lambda) = \exp\left(-\frac{i}{4}\omega_{\mu\nu}\sigma^{\mu\nu}\right). \quad (165)$$

Consistency condition:

$$S^{-1}(\Lambda)\gamma^\mu S(\Lambda) = \Lambda^\mu_\nu\gamma^\nu. \quad (166)$$

### 5.4 The Dirac Equation and Dirac Lagrangian

The Dirac equation:

$$(i\gamma^\mu\partial_\mu - m)\psi = 0. \quad (167)$$

Dirac Lagrangian:

$$\mathcal{L}_D = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi, \quad \bar{\psi} := \psi^\dagger\gamma^0. \quad (168)$$

Euler-Lagrange gives

$$(i\gamma^\mu\partial_\mu - m)\psi = 0, \quad \bar{\psi}(i\overleftarrow{\partial}_\mu\gamma^\mu + m) = 0. \quad (169)$$

**Klein–Gordon relation.** Multiply the Dirac operator:

$$(i\gamma^\mu \partial_\mu - m)(i\gamma^\nu \partial_\nu + m)\psi = (-\partial^2 - m^2)\psi = 0, \quad (170)$$

so each component satisfies the KG equation.

### 5.5 Chiral Projectors and Chiral Spinors

Define projectors:

$$P_L = \frac{1 - \gamma^5}{2}, \quad P_R = \frac{1 + \gamma^5}{2}. \quad (171)$$

Decompose:

$$\psi = \psi_L + \psi_R, \quad \psi_{L,R} = P_{L,R}\psi. \quad (172)$$

For  $m \neq 0$ , left and right components mix in the Dirac equation. For  $m = 0$ , they decouple.

### 5.6 The Weyl Equation (Massless Spinors)

For  $m = 0$ :

$$i\gamma^\mu \partial_\mu \psi = 0. \quad (173)$$

Chiral components satisfy separate Weyl equations:

$$i\gamma^\mu \partial_\mu \psi_L = 0, \quad i\gamma^\mu \partial_\mu \psi_R = 0. \quad (174)$$

Interpretation: massless fermions can have definite chirality.

### 5.7 Parity

Parity acts on coordinates:

$$P : (t, \mathbf{x}) \rightarrow (t, -\mathbf{x}). \quad (175)$$

On spinors (standard convention):

$$\psi(t, \mathbf{x}) \xrightarrow{P} \psi'(t, \mathbf{x}) = \gamma^0 \psi(t, -\mathbf{x}). \quad (176)$$

Under parity, chirality flips:

$$\psi_L \leftrightarrow \psi_R. \quad (177)$$

Thus parity is incompatible with purely chiral theories (e.g. weak interaction).

### 5.8 Majorana Spinors (Idea)

A Majorana spinor satisfies a reality condition:

$$\psi^c = \psi, \quad (178)$$

where

$$\psi^c := C\bar{\psi}^T, \quad (179)$$

and  $C$  is the charge conjugation matrix (representation-dependent).

**Physical meaning.** A Majorana fermion is its own antiparticle. Majorana mass terms schematically look like

$$\mathcal{L}_M \sim -\frac{m}{2} (\bar{\psi}\psi^c + \bar{\psi}^c\psi), \quad (180)$$

and require neutral fermions (no conserved  $U(1)$  charge).

## 5.9 Symmetries and Conserved Currents

**Global  $U(1)$  phase symmetry.**

$$\psi \rightarrow e^{i\alpha}\psi, \quad \bar{\psi} \rightarrow \bar{\psi}e^{-i\alpha}. \quad (181)$$

Noether current:

$$j^\mu = \bar{\psi}\gamma^\mu\psi, \quad \partial_\mu j^\mu = 0. \quad (182)$$

Conserved charge:

$$Q = \int d^3x j^0. \quad (183)$$

**Axial (chiral) current.** For  $m = 0$ , the Lagrangian is also invariant under

$$\psi \rightarrow e^{i\alpha\gamma^5}\psi, \quad (184)$$

with axial current

$$j_5^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi. \quad (185)$$

Classically,

$$\partial_\mu j_5^\mu = 0 \quad (m = 0), \quad (186)$$

but quantum mechanically this symmetry can be broken by anomalies (important in gauge theories).

## 5.10 Plane Wave Solutions

Seek solutions of the form

$$\psi(x) = u(p)e^{-ip \cdot x}, \quad p^0 = E_{\mathbf{p}}. \quad (187)$$

Plugging into Dirac equation gives

$$(\not{p} - m)u(p) = 0, \quad \not{p} := \gamma^\mu p_\mu. \quad (188)$$

Similarly for negative-frequency solutions:

$$\psi(x) = v(p)e^{ip \cdot x}, \quad (\not{p} + m)v(p) = 0. \quad (189)$$

**Spin sums (very useful).**

$$\sum_s u^{(s)}(p)\bar{u}^{(s)}(p) = \not{p} + m, \quad \sum_s v^{(s)}(p)\bar{v}^{(s)}(p) = \not{p} - m. \quad (190)$$

**Normalization (common covariant choice).**

$$\bar{u}^{(s)}(p)u^{(r)}(p) = 2m\delta^{sr}, \quad \bar{v}^{(s)}(p)v^{(r)}(p) = -2m\delta^{sr}. \quad (191)$$

**5.11 Summary (Exam Checklist)**

- Gamma matrices satisfy  $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ .
- Dirac equation:  $(i\not{\partial} - m)\psi = 0$ .
- Lorentz transformations on spinors:  $S(\Lambda) = \exp(-\frac{i}{4}\omega_{\mu\nu}\sigma^{\mu\nu})$ .
- Chirality projectors:  $P_{L,R} = (1 \mp \gamma^5)/2$ .
- Weyl spinors: massless chiral components decouple.
- Parity:  $\psi(t, \mathbf{x}) \rightarrow \gamma^0\psi(t, -\mathbf{x})$ , flips chirality.
- Conserved current:  $j^\mu = \bar{\psi}\gamma^\mu\psi$  from global  $U(1)$ .
- Plane waves:  $(\not{p} \mp m)u, v = 0$  and spin sums  $\sum u\bar{u} = \not{p} + m$ .

## 6 Quantizing the Dirac Field

### 6.1 A Glimpse at the Spin–Statistics Theorem (What You Need)

Empirical + theoretical result:

$$\text{integer spin} \Rightarrow \text{bosons (commutators)}, \quad \text{half-integer spin} \Rightarrow \text{fermions (anticommutators)}. \quad (192)$$

Why it matters (minimal statement):

- Using commutators for spin- $\frac{1}{2}$  fields leads to negative-norm states or violation of causality.
- Using anticommutators ensures positivity of energy and consistent relativistic causality.

### 6.2 Canonical Fermionic Quantization

Start from the Dirac Lagrangian:

$$\mathcal{L}_D = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi. \quad (193)$$

Canonical momentum conjugate to  $\psi$ :

$$\pi_\psi := \frac{\partial \mathcal{L}}{\partial(\partial_0 \psi)} = i\psi^\dagger. \quad (194)$$

Impose equal-time **anticommutation relations**:

$$\{\psi_\alpha(\mathbf{x}, t), \psi_\beta^\dagger(\mathbf{y}, t)\} = \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}), \quad (195)$$

$$\{\psi_\alpha(\mathbf{x}, t), \psi_\beta(\mathbf{y}, t)\} = 0, \quad \{\psi_\alpha^\dagger(\mathbf{x}, t), \psi_\beta^\dagger(\mathbf{y}, t)\} = 0. \quad (196)$$

Here  $\alpha, \beta$  are spinor indices  $(1, \dots, 4)$ .

### 6.3 Mode Expansion of the Dirac Field

The free Dirac field operator is expanded as

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \sum_{s=1,2} \left( b_{\mathbf{p}}^{(s)} u^{(s)}(p) e^{-ip \cdot x} + d_{\mathbf{p}}^{(s)\dagger} v^{(s)}(p) e^{ip \cdot x} \right), \quad p^0 = E_{\mathbf{p}}. \quad (197)$$

Interpretation:

- $b_{\mathbf{p}}^{(s)}$  annihilates a particle (fermion).
- $d_{\mathbf{p}}^{(s)}$  annihilates an antiparticle (antifermion).

Anticommutators:

$$\{b_{\mathbf{p}}^{(r)}, b_{\mathbf{q}}^{(s)\dagger}\} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}) \delta^{rs}, \quad \{d_{\mathbf{p}}^{(r)}, d_{\mathbf{q}}^{(s)\dagger}\} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}) \delta^{rs}, \quad (198)$$

all others vanish.

## 6.4 Fermi–Dirac Statistics and the Pauli Principle

Because operators anticommute,

$$\left(b_{\mathbf{p}}^{(s)\dagger}\right)^2 = 0, \quad \left(d_{\mathbf{p}}^{(s)\dagger}\right)^2 = 0, \quad (199)$$

so no two identical fermions can occupy the same one-particle state:

$$n_{\mathbf{p}}^{(s)} \in \{0, 1\}. \quad (200)$$

This is the Pauli exclusion principle.

Number operators:

$$N_b = \int \frac{d^3p}{(2\pi)^3} \sum_s b_{\mathbf{p}}^{(s)\dagger} b_{\mathbf{p}}^{(s)}, \quad N_d = \int \frac{d^3p}{(2\pi)^3} \sum_s d_{\mathbf{p}}^{(s)\dagger} d_{\mathbf{p}}^{(s)}. \quad (201)$$

## 6.5 Hamiltonian and Energy Positivity

The free Dirac Hamiltonian can be written (after normal ordering) as

$$: H := \int \frac{d^3p}{(2\pi)^3} \sum_s E_{\mathbf{p}} \left( b_{\mathbf{p}}^{(s)\dagger} b_{\mathbf{p}}^{(s)} + d_{\mathbf{p}}^{(s)\dagger} d_{\mathbf{p}}^{(s)} \right). \quad (202)$$

Hence both particles and antiparticles contribute positive energy.

## 6.6 Particles and Antiparticles

Define vacuum:

$$b_{\mathbf{p}}^{(s)}|0\rangle = 0, \quad d_{\mathbf{p}}^{(s)}|0\rangle = 0 \quad \forall \mathbf{p}, s. \quad (203)$$

One-particle states:

$$|\mathbf{p}, s\rangle = b_{\mathbf{p}}^{(s)\dagger}|0\rangle, \quad |\overline{\mathbf{p}}, \overline{s}\rangle = d_{\mathbf{p}}^{(s)\dagger}|0\rangle. \quad (204)$$

Conserved  $U(1)$  current:

$$j^\mu = \bar{\psi} \gamma^\mu \psi, \quad Q = \int d^3x j^0. \quad (205)$$

In terms of operators (schematically):

$$Q = \int \frac{d^3p}{(2\pi)^3} \sum_s \left( b_{\mathbf{p}}^{(s)\dagger} b_{\mathbf{p}}^{(s)} - d_{\mathbf{p}}^{(s)\dagger} d_{\mathbf{p}}^{(s)} \right). \quad (206)$$

Thus antiparticles carry opposite charge.

## 6.7 Dirac's Hole Interpretation (Historical Motivation)

The Dirac equation has negative-energy solutions. Dirac proposed:

- All negative-energy states are filled in the vacuum (Dirac sea).
- A missing negative-energy electron appears as a positively charged particle (the positron).

Modern QFT interpretation: negative-energy modes are reinterpreted as antiparticles with positive energy.

## 6.8 Fermion Propagator

The **Feynman propagator** for the Dirac field:

$$S_F(x - y) := \langle 0 | T \{ \psi(x) \bar{\psi}(y) \} | 0 \rangle. \quad (207)$$

Momentum-space representation:

$$S_F(x - y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon} e^{-ip \cdot (x - y)}. \quad (208)$$

It satisfies the Green function equation:

$$(i\not{\partial}_x - m) S_F(x - y) = i\delta^{(4)}(x - y). \quad (209)$$

## 6.9 Spin Sums (Used Constantly)

$$\sum_s u^{(s)}(p) \bar{u}^{(s)}(p) = \not{p} + m, \quad \sum_s v^{(s)}(p) \bar{v}^{(s)}(p) = \not{p} - m. \quad (210)$$

## 6.10 Wick's Theorem for Fermions (Sign Rules)

Wick's theorem also holds for fermions, but each exchange of fermionic operators introduces a minus sign.

Fermion contraction:

$$\overline{\psi(x) \bar{\psi}(y)} = \langle 0 | T \{ \psi(x) \bar{\psi}(y) \} | 0 \rangle = S_F(x - y). \quad (211)$$

Time ordering for fermions includes signs:

$$T \{ \psi(x) \bar{\psi}(y) \} = \theta(x^0 - y^0) \psi(x) \bar{\psi}(y) - \theta(y^0 - x^0) \bar{\psi}(y) \psi(x). \quad (212)$$

## 6.11 Feynman Rules (Dirac Fermions, Minimal Set)

For a theory with interaction  $\mathcal{L}_{\text{int}}$ :

$$S = T \exp \left( i \int d^4 x \mathcal{L}_{\text{int}}(x) \right). \quad (213)$$

**Free fermion line (propagator).** Internal fermion line with momentum  $p$ :

$$\frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}. \quad (214)$$

**External spinors.** For external fermion legs:

$$\text{incoming fermion: } u^{(s)}(p), \quad \text{outgoing fermion: } \bar{u}^{(s)}(p), \quad (215)$$

$$\text{incoming antifermion: } \bar{v}^{(s)}(p), \quad \text{outgoing antifermion: } v^{(s)}(p). \quad (216)$$

**Closed fermion loops.** Each closed fermion loop contributes an extra factor:

$$(-1). \quad (217)$$

**Example: Yukawa vertex.** If

$$\mathcal{L}_{\text{int}} = -g \bar{\psi} \psi \phi, \quad (218)$$

then the vertex factor is

$$-ig. \quad (219)$$

**Example: QED vertex.** If

$$\mathcal{L}_{\text{int}} = -e \bar{\psi} \gamma^\mu \psi A_\mu, \quad (220)$$

then vertex factor:

$$-ie\gamma^\mu. \quad (221)$$

## 6.12 Summary (Exam Checklist)

- Fermions must be quantized with anticommutators:

$$\{\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})\} = \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

- Mode expansion introduces particle operators  $b$  and antiparticle operators  $d$ .
- Pauli principle:  $(b^\dagger)^2 = 0$  and occupancy  $n \in \{0, 1\}$ .
- Dirac propagator:

$$S_F(p) = \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}.$$

- Spin sums:

$$\sum u \bar{u} = \not{p} + m, \quad \sum v \bar{v} = \not{p} - m.$$

- Closed fermion loop  $\Rightarrow$  extra factor  $(-1)$ .

## 7 Quantum Electrodynamics (QED)

### 7.1 Gauge Invariance (Classical EM)

The electromagnetic field is described by a 4-potential  $A_\mu$  with field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (222)$$

Gauge transformation:

$$A_\mu(x) \rightarrow A'_\mu(x) = A_\mu(x) + \partial_\mu \alpha(x). \quad (223)$$

Since  $F_{\mu\nu}$  is invariant, Maxwell theory can be written as

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \quad (224)$$

### 7.2 Inclusion of Matter: Minimal Coupling (QED Lagrangian)

Start from the free Dirac Lagrangian and impose *local*  $U(1)$  invariance:

$$\psi(x) \rightarrow e^{ie\alpha(x)} \psi(x). \quad (225)$$

Introduce the covariant derivative

$$D_\mu := \partial_\mu + ieA_\mu, \quad (226)$$

which transforms covariantly:

$$D_\mu \psi \rightarrow e^{ie\alpha(x)} D_\mu \psi. \quad (227)$$

The full QED Lagrangian:

$$\mathcal{L}_{\text{QED}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \quad (228)$$

Expanding:

$$\mathcal{L}_{\text{QED}} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - e \bar{\psi} \gamma^\mu \psi A_\mu. \quad (229)$$

The interaction term is:

$$\mathcal{L}_{\text{int}} = -e \bar{\psi} \gamma^\mu \psi A_\mu. \quad (230)$$

### 7.3 Conserved Current

Gauge invariance implies a conserved  $U(1)$  current:

$$j^\mu = \bar{\psi} \gamma^\mu \psi, \quad \partial_\mu j^\mu = 0, \quad (231)$$

and Maxwell's equation becomes

$$\partial_\mu F^{\mu\nu} = e j^\nu. \quad (232)$$

## 7.4 Quantization of the EM Field (Key Issue)

A massless spin-1 field has gauge redundancy, so naive canonical quantization is subtle. The physical photon has only **two** transverse polarizations.

To quantize covariantly, introduce a gauge-fixing term. In Lorenz gauge:

$$\partial_\mu A^\mu = 0. \quad (233)$$

A standard gauge-fixed Lagrangian (covariant gauges):

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2. \quad (234)$$

Here  $\xi$  is the gauge parameter. The most common choice:

$$\xi = 1 \quad (\text{Feynman gauge}). \quad (235)$$

## 7.5 Lorentz-Invariant Propagators

**Fermion propagator.**

$$S_F(p) = \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}. \quad (236)$$

**Photon propagator (covariant gauge).** In momentum space:

$$D_{\mu\nu}(k) = \frac{-i}{k^2 + i\epsilon} \left( \eta_{\mu\nu} - (1 - \xi) \frac{k_\mu k_\nu}{k^2} \right). \quad (237)$$

In Feynman gauge ( $\xi = 1$ ):

$$D_{\mu\nu}(k) = \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}. \quad (238)$$

## 7.6 QED Feynman Rules (Momentum Space)

From

$$\mathcal{L}_{\text{int}} = -e \bar{\psi} \gamma^\mu \psi A_\mu, \quad (239)$$

the vertex factor is:

$$-ie\gamma^\mu. \quad (240)$$

**Summary of rules.**

- Fermion propagator:

$$\frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}.$$

- Photon propagator (Feynman gauge):

$$\frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

- Vertex:

$$-ie\gamma^\mu.$$

- External fermion legs:

$$u^{(s)}(p), \bar{u}^{(s)}(p), v^{(s)}(p), \bar{v}^{(s)}(p).$$

- Momentum conservation at each vertex:

$$(2\pi)^4 \delta^{(4)}\left(\sum p_{\text{in}} - \sum p_{\text{out}}\right).$$

- Closed fermion loop: extra factor  $(-1)$ .

## 7.7 QED Processes (Core Examples)

### 7.7.1 Electron–Muon Scattering: $e^- \mu^- \rightarrow e^- \mu^-$

Tree-level diagram: single photon exchange ( $t$ -channel).

Invariant amplitude:

$$i\mathcal{M} = \bar{u}(p_3)(-ie\gamma^\mu)u(p_1) \left( \frac{-i\eta_{\mu\nu}}{(p_1 - p_3)^2 + i\epsilon} \right) \bar{u}(p_4)(-ie\gamma^\nu)u(p_2). \quad (241)$$

So

$$\mathcal{M} = (-ie)^2 [\bar{u}(p_3)\gamma^\mu u(p_1)] \frac{\eta_{\mu\nu}}{q^2} [\bar{u}(p_4)\gamma^\nu u(p_2)], \quad q = p_1 - p_3. \quad (242)$$

### 7.7.2 $e^+ e^- \rightarrow \mu^+ \mu^-$ (Annihilation)

Tree-level diagram:  $s$ -channel photon exchange.

Amplitude:

$$i\mathcal{M} = \bar{v}(p_2)(-ie\gamma^\mu)u(p_1) \left( \frac{-i\eta_{\mu\nu}}{(p_1 + p_2)^2 + i\epsilon} \right) \bar{u}(p_3)(-ie\gamma^\nu)v(p_4). \quad (243)$$

### 7.7.3 Compton Scattering: $\gamma e^- \rightarrow \gamma e^-$

Two tree-level diagrams:

- $s$ -channel: intermediate electron after absorbing photon
- $u$ -channel: intermediate electron after emitting photon

Key point: both are required by gauge invariance.

### 7.7.4 Møller and Bhabha Scattering

- Møller:  $e^- e^- \rightarrow e^- e^-$  (exchange + identical fermion effects)
- Bhabha:  $e^- e^+ \rightarrow e^- e^+$  (both  $s$  and  $t$  channels)

## 7.8 Gauge Invariance and Ward Identity (Essential Idea)

Gauge invariance implies relations between amplitudes. A key practical consequence: physical amplitudes are unchanged under

$$\epsilon_\mu(k) \rightarrow \epsilon_\mu(k) + c k_\mu \quad (244)$$

for external photons (with polarization vector  $\epsilon_\mu$ ).

This leads to the Ward identity (schematically):

$$k_\mu \mathcal{M}^\mu = 0. \quad (245)$$

This ensures consistency of the theory and cancellation of unphysical polarizations.

## 7.9 Summary (Exam Checklist)

- QED Lagrangian:

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}.$$

- Gauge transformation:

$$A_\mu \rightarrow A_\mu + \partial_\mu \alpha, \quad \psi \rightarrow e^{ie\alpha}\psi.$$

- Interaction vertex:

$$-ie\gamma^\mu.$$

- Propagators:

$$S_F(p) = \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}, \quad D_{\mu\nu}(k) = \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon} \quad (\xi = 1).$$

- Typical tree-level processes: scattering via photon exchange, annihilation, Compton scattering.
- Ward identity expresses gauge invariance: demonstrate  $k_\mu \mathcal{M}^\mu = 0$ .